

UNDERSTANDING DNA

A SELF-COPYING MOLECULE CALLED DNA IS THE CHEMICAL CODE OF LIFE ITSELF

The characteristics of living things are produced by chemical processes that happen in every cell. Twentieth-century science traced these processes to their source—a molecule that not only carries genetic information but also has the remarkable capacity to copy itself. It is called DNA.

By the turn of the 20th century, scientists had discovered that inherited characteristics come from particles passed down through generations. However, they did not understand the composition of these units of genetic material, which we now know as genes. In 1919, Lithuanian biochemist Phoebus Levene dismissed nucleic acid—a material present in the nucleus of every cell—as too simple to be involved directly in inheritance. But in the decades that followed, experiments proved that a form of nucleic acid is, in fact, the substance of genes.

By the 1950s, advances in analytical techniques meant that the best-known form of nucleic acid, DNA (deoxyribonucleic acid), could be examined in ways never before possible. A method called X-ray crystallography even promised to reveal its three-dimensional shape.

BASE PAIRS

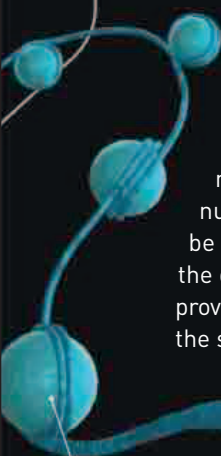
In 1953, results from these new methods convinced American biologist James Watson and British biophysicist Francis Crick that DNA had a helical structure. Evidence indicated two coiled chains (a double helix) with variable components (bases) that hold the chains together. Four varieties of bases were always present in certain proportions. Watson and Crick deduced that this



JAMES WATSON AND FRANCIS CRICK

To test their double helix theory, Watson and Crick built a model of the structure to check that the chemical pieces would fit.

was because they were bonded in fixed ways: adenine with thymine and guanine with cytosine. This would be key to understanding not only how DNA carried inherited information but also how this information replicated at reproduction.



double helix wrapped around packaging proteins called histones

DNA molecule coiled into a double helix

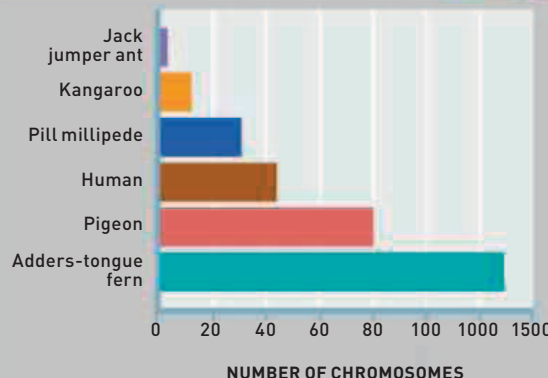
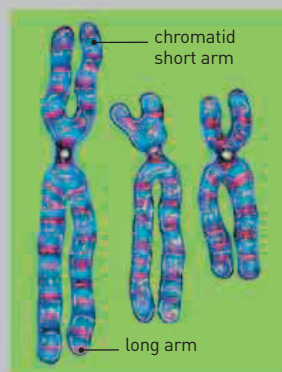
adenine
thymine

CHROMOSOMES

When a cell divides, to prevent entanglement each DNA molecule is bundled into condensed structures called chromosomes. The number of chromosomes per cell varies from species to species.

Chromosome micrograph

Because DNA replication happens before cell division, every chromosome appears with duplicated "chromatids."



DNA backbone, consisting of deoxyribose (a form of sugar) and phosphates